Exercises Comprehensions

Exercise

Let's say I give you a list saved in a variable: a = [1, 4, 9, 16, 25, 36, 49, 64, 81, 100]. Write one line of Python that takes this list a and makes a new list that has only the even elements of this list in it.

```
# List comprehension 1

a = [1, 4, 9, 16, 25, 36, 49, 64, 81, 100]
b = [number for number in a if number % 2 == 0]
print (b)
```

Exercise

Find all x where x is a natural number less than or equal to 100 and x is a perfect square.
This can be solved using a for-loop as :

```
for i in range(1,101):     #the iterator
   if int(i**0.5)==i**0.5: #conditional filtering
     print i               #output-expression
```

Now create a list comprehension doing the same

```
# List comprehension 2

li1 = [i for i in range(1,101) if int(i**0.5)==i**0.5]
print (li1)
```

Exercise

Take a matrix as input and return a list with each row placed on after the other.
The Python code with a FOR-loop looks as follows :

```
def eg1_for(matrix):
    flat = []
    for row in matrix:
        for x in row:
            flat.append(x)
    return flat
```

Let's define a matrix and test the results:

```
matrix = [ range(0,5), range(5,10), range(10,15) ]
print "Original Matrix: " + str(matrix)
print "FOR-loop result: " + str(eg1_for(matrix))
```

Now create the List Comprehension that does the same.

```
# List comprehension 3

def eg1_lc(matrix):
    return [x for row in matrix for x in row ]

matrix = [ range(0,5), range(5,10), range(10,15) ]
print ("Original Matrix: " + str(matrix))
print ("LC result      : " + str(eg1_lc(matrix)))
```

Exercise

Take two list of same length as input and return a dictionary with one as keys and other as values. The Python codes with FOR-loop looks as follows :

```
def eg3_for(keys, values):
    dic = {}
    for i in range(len(keys)):
        dic[keys[i]] = values[i]
    return dic
```

Define a matrix and test the results :

```
country = ['Germany', 'France', 'Belgium', 'England', 'Spain', 'Italy']
capital = ['Berlin', 'Paris','Brussels', 'London', 'Madrid', 'Rome']
print "FOR-loop result: " + str(eg3_for(country, capital))
```

Now create a solution using dictionary comprehension.

```
# Dictionary comprehension 4

def eg3_lc(keys, values):
    return { keys[i] : values[i] for i in range(len(keys)) }

country = ['Germany', 'France', 'Belgium', 'England', 'Spain', 'Italy']
capital = ['Berlin', 'Paris','Brussels', 'London', 'Madrid', 'Rome']
print ("LC result      : " + str(eg3_lc(country, capital)))
```